

E-commerce Sales Data Cleaning

Data Quality Audit & Cleaning Case Study — Excel Formulas + Power Query

Author: Franklin Manuel Ruiz Guadamuz · Tools: Excel, Power Query · Source: Kaggle — messy e-commerce sales export (103 records)

1. OBJECTIVE

Audit and clean a raw e-commerce sales export so it can be used reliably for reporting, while documenting every issue found and every decision taken. The project was delivered in two independent versions — one built entirely with formulas (transparent and auditable) and one built as a Power Query pipeline (automated and refreshable) — to compare both approaches and cross-validate the results.

2. SOURCE DATA

Attribute	Detail
Records in / out	103 → 100
Columns	11 (ID, Customer, Order ID, Date, Product, Category, Quantity, Price, Payment, Status, Total)
Issues catalogued	19 distinct data quality issues
Deliverable	Cleaned dataset + audit log + data quality flag column

3. METHOD

The workbook is organised so the work is fully traceable: **01_Raw** preserves the untouched source; **02_Audit** logs each issue with severity and the decision taken; **03_Clean_Formulas** shows raw and cleaned values side by side so every transformation can be inspected; **04_Clean_Final** is the client-facing deliverable in values only; and **05_Clean_PowerQuery** holds the automated pipeline output.

ID	CustomerName	OrderID	OrderDate	Product	Category	Quantity	Price	Payment	Status	Total
001	Customer 001	1000-0001-10000001	2020-01-01	Smart Watch	Electronics	1	25	PayPal	Completed	25.00
002	Customer 001	1000-0001-10000002	2020-01-01	Smart Watch	Electronics	2	25.00	PayPal	Cancelled	50.00
003	Customer 002	1000-0002-10000001	2020-01-02	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
004	Customer 004	1000-0004-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
005	Customer 005	1000-0005-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
006	Customer 006	1000-0006-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
007	Customer 007	1000-0007-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
008	Customer 008	1000-0008-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
009	Customer 009	1000-0009-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
010	Customer 010	1000-0010-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
011	Customer 011	1000-0011-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
012	Customer 012	1000-0012-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
013	Customer 013	1000-0013-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
014	Customer 014	1000-0014-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
015	Customer 015	1000-0015-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
016	Customer 016	1000-0016-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
017	Customer 017	1000-0017-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
018	Customer 018	1000-0018-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
019	Customer 019	1000-0019-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
020	Customer 020	1000-0020-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
021	Customer 021	1000-0021-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
022	Customer 022	1000-0022-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
023	Customer 023	1000-0023-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
024	Customer 024	1000-0024-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
025	Customer 025	1000-0025-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
026	Customer 026	1000-0026-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
027	Customer 027	1000-0027-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
028	Customer 028	1000-0028-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
029	Customer 029	1000-0029-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
030	Customer 030	1000-0030-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
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032	Customer 032	1000-0032-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
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039	Customer 039	1000-0039-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
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070	Customer 070	1000-0070-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
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081	Customer 081	1000-0081-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
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089	Customer 089	1000-0089-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
090	Customer 090	1000-0090-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
091	Customer 091	1000-0091-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
092	Customer 092	1000-0092-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
093	Customer 093	1000-0093-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
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097	Customer 097	1000-0097-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
098	Customer 098	1000-0098-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
099	Customer 099	1000-0099-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00
100	Customer 100	1000-0100-10000001	2020-01-03	Smart Watch	Electronics	1	25.00	PayPal	Pending	25.00

Before — raw export: mixed date formats, text in numeric fields, currency symbols, inconsistent casing, hidden nulls

After — final deliverable: standardized dates and categories, numeric fields, and a per-record data quality flag

Critical — silent date misparsing

Critical — duplicates with conflicting values

Critical — an undocumented discount pattern

High — hidden and impossible values

- ### Medium / Low — standardization and recovery

- ## 5. GUIDING PRINCIPLES

- **Never delete client data silently.** Missing and invalid values are labeled, not removed.
- **Missing $\neq$ invalid.** An empty cell and corrupted text are different problems requiring different fixes.
- **Flag, don't guess.** Outliers and impossible values go back to the client for confirmation.
- **Validate against business logic, not just data types.** A negative quantity passes every type check and is still wrong.

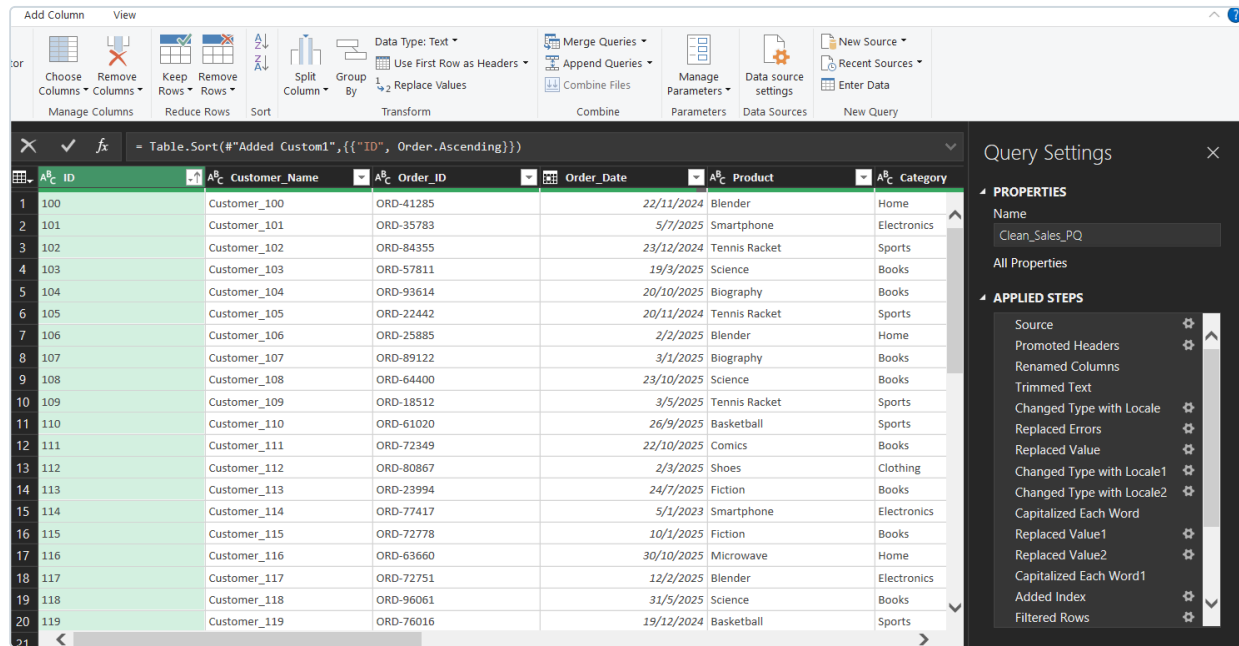
Audit log — 19 issues catalogued with severity and the decision taken for each

Task	Formulas	Power Query
Date parsing (locale)	Nested FIND / LEFT / MID / RIGHT / DATE	One locale-aware type change
Decimal separator	SUBSTITUTE in every numeric column	Handled in the same step
Trim all text	Column by column	One step, all columns
Fixing a mistake	Rework, risk of row misalignment	Delete the step
Reusable next month	Rebuild from scratch	Refresh

Automation executes decisions; it does not make them.

Formula version — raw and cleaned values side by side, keeping every transformation auditable

Formula version — raw and cleaned values side by side, keeping every transformation auditable



Power Query version — the same result as a sequence of reversible, refreshable applied steps

7. RESULTS

Metric	Value
Records in / out	103 → 100
Duplicates resolved	3 (one exact, two validated against business logic)
Issues catalogued	19 (3 critical, 4 high, 8 medium, 4 low)
Category values standardized	9 → 5
Records flagged for client review	3 discrepancies + 4 impossible values
Fields recovered	1 Price, 1 Total

Deliverables: cleaned dataset (values only, with a data-quality flag per record), a full audit log with severity and decisions, a formula-based version for transparency, and a Power Query pipeline that can be refreshed against future files.